

A COMPARATIVE ANALYSIS OF SELF-PERCEPTION: GIFTED VS. NON-GIFTED BACKGROUND

Leia Mayssa binti Masyerin^{1*}, Qawiemah Batrisyia binti Syaiful Izzat¹,
Aleesa Sofea binti M. Fadzil¹, Noralia binti Arman¹, Nurul Suzaina binti Joli¹,
Che Suriani binti Kiflee¹

¹Kolej PERMATA@Pintar Negara,
Universiti Kebangsaan Malaysia,
43600 UKM Bangi, Selangor, Malaysia.

**Corresponding author: m001262@geniuspintar.ukm.edu.my*

ABSTRACT

This research study examines the differences in self-perception across giftedness and gender, with the main emphasis on giftedness. This research evaluates a wide range of aspects of self-perception, including intellectual abilities, social skills and personal attributes. A questionnaire, adapted from existing instruments such as The Piers-Harris Self-Concept Scale (2002) and Self Description Questionnaire-I (1990), was conducted, in which over 300 responses were collected from Malaysian individuals of varying backgrounds. Descriptive statistics and independent sample t-tests were then performed using SPSS (Statistical Package for the Social Sciences) software, with a 95% confidence level ($\alpha = 0.05$), to assess group differences. The analysis revealed that non-gifted individuals generally have higher levels of self-perception (overall mean of 4.60) compared to gifted individuals (overall mean of 4.44) with a significance level of $\alpha = 0.010$. Moreover, gifted individuals showed higher perceptions of their IQ, intelligence, language skills, musical abilities, and artistic abilities, while non-gifted individuals showed higher perceptions of their science proficiency and overall happiness. However, gender had little overall effect on self-perception, excluding physical activeness and musical abilities, in which females indicated higher ratings than males. Furthermore, most of the graphs obtained illustrated normal distributions with a rightward shift, indicating a tendency for individuals to perceive themselves more positively than may be objectively warranted. The findings of this study contribute to a better understanding of how an individual's giftedness and gender affect self-perception in Malaysia. Further exploration and analysis of the survey data will shed light on the intricate dynamics of self-perception and its implications for individuals from different backgrounds.

Keywords: *self-perception; self-concept; perception; giftedness; gifted vs non-gifted*

1.0 Introduction

Self-perception is defined as a person's view of themselves, or any of the mental or physical attributes that constitute the self. It is not entirely accurate to reality as it may experience varying degrees of distortion (APA Dictionary of Psychology, 2007). Giftedness is identified by high ability, creativity and task commitment (Renzulli, J.S, 2011) . Previous research has shown that gifted students generally tend to have higher self-perception than students that are not labelled as gifted, although some studies show the opposite (Albright, N.M., 1965; Hoge, R.D. & Renzulli, J.S., 1993). Gender also has an effect on self-perception due to the different upbringing and priorities of females and males. Yet, little to no research has been done on the effects of gender and giftedness on self-perception in the context of gifted education in Malaysia.

Thus, this study has been conducted with the following objectives:

1. To collect and analyse the self-perception of gifted individuals and the general public.
2. To study the differences in self-perception between genders.
3. To study the differences in self-perception between gifted and non-gifted individuals in Malaysia.

Two hypotheses based on these objectives have been formulated:

H_{01} = There is no significant difference in self-perception between gifted and non-gifted individuals.

H_{a1} = There is a significant difference in self-perception between gifted and non-gifted individuals.

H_{02} = There is no significant difference in self-perception between males and females.

H_{a2} = There is a significant difference in self-perception between males and females.

2.0 Literature Review

2.1 Self-Perception

Self-perception is an individual's personal view of the physical and mental qualities that define the self (APA Dictionary of Psychology, 2007). People, on average, tend to provide more positive ratings of themselves than what is evaluated by others (Robins & John, 1994). This same data set also showed that individuals who overestimated their performance ratings had higher associations with maladjustment compared to those who underestimated their ratings (Robins & John, 1997).

2.2 Giftedness in Malaysia

The study of giftedness comprises countless opposing theories and definitions due to the unclear nature of what is looked for in 'gifted' individuals (Feldhusen & Keller, 1986). In general, it is characterised by increased general ability, creativity, commitment to tasks, higher-order processes and asynchronous development (Renzulli, J.S., 2011; Sternberg, R.J.,

1981; Silverman, L.K., 1997). In Malaysia, formal evaluations of giftedness are conducted by Pusat PERMATApintar. The organisation analyses various components utilising UKM 1 and 2 tests (intellectual), Torrance Tests of Creative Thinking (creativity) and Malaysian Emotional Quotient Inventory (socioaffective) via an online platform. A three-week school holiday programme was also conducted to evaluate sensorimotor abilities, leadership and scientific acumen through observation (Abu Bakar & Mohd. Ishak, 2017).

2.3 The Effect of Gender and Giftedness on Self-Perception

Due to the different social and biological conditions of their upbringing, we found gender to be another point of interest. Gender has an effect on the different mechanisms affecting self-perception. In an academic setting, female students are more affected by personal and direct negative feedback (i.e. failing a class). On the other hand, male students are more influenced by status and labels (i.e. being diagnosed with a learning disability) (Crosnoe, R., Riegle-Crumb, C. & Muller, C., 2007). Female students also exhibit higher confidence in their language abilities compared to males (Watt, H.M.G., 2004) although this differs among articles (Jacobs, J.E., Lanza, S., Osgood, D.W., Eccles, J.S., & Wigfield, A., 2002). Overall, most gender-based comparisons fail to yield significant results (Hoge, R.D. & Renzulli, J.S., 1993).

Existing literature on the self-perception of gifted individuals often displays contradicting results, possibly due to a lack of significant results or insufficient sample sizes (Hoge, R.D. & Renzulli, J.S., 1993). Gifted students typically have higher academic but lower physical and athletic perceptions (Albright, N.M., 1965; Hoge, R.D. & Renzulli, J.S., 1993). Despite this, the same studies conflict on behaviour and conduct perception. Overall, most research indicates that gifted individuals tend to have higher self-perception. In spite of that, differences in self-perception may change or even switch completely over time (Yan, K. & Haihui, Z., 2005). However, placing them in special programmes has a negative effect on their self-perception (Hoge, R.D. & Renzulli, J.S., 1993). Theory asserts that placing gifted students with similarly intelligent peers will cause a more negative self-perception as the benchmark for ‘talent’ shifts (Yan, K. & Haihui, Z., 2005). A positive self-perception is important for gifted students as it indicates greater proficiency of tasks and more complex learning strategies. (Dean, R.S., 1977)

3.0 Methodology

3.1 Questionnaire

This research utilised a questionnaire conducted via Google Form in order to study self-perception. The questionnaire was developed by drawing inspiration from existing instruments and adapting them to this study. Table 1 is a list of instruments and its respective items that were used as reference.

Table 1. Reviewed Instruments and Related Items

Instrument	Item
The Piers-Harris Self-Concept Scale (2002)	- Intellectual and school status - Popularity - Physical appearance and attributes - Happiness and satisfaction

Self-Perception Profile for Children (1985)	• Scholastic Competence • Social Competence	• Athletic Competence • Physical Appearance
Self Description Questionnaire-I (1990)	• Physical Ability • Physical Appearance • Peer Relations	• Reading • Mathematics • General-School
The Tennessee Self Concept Scale (1965)	• Physical self-concept • Social self-concept • Emotional self-concept • Academic self-concept • Moral-ethical self-concept	• Religious self-concept • Personal self-concept • Interpersonal self-concept • General self-concept

The questionnaire contained two sections:

1. Demographics: Participants' gender and giftedness information were collected in this section. For this research's scope, individuals falling under the category of "gifted" include those who are either students or alumni of gifted institutions (i.e. PERMATA@Pintar, PERMATA Insan and Akademi Sains Pendang), while all others are categorised as "non-gifted."

2. Survey on Self-Perception: There were a total of 40 questions, each related to a different trait/ability as shown in Table 2. For each question, participants were required to rank themselves (based on their own self-perception) in terms of that aspect in comparison to other Malaysians in their age group. There were 7 ranking options resembling a likert scale i.e. (1) way below average, (2) below average, (3) slightly below average, (4) average, (5) slightly above average, (6) above average, and (7) way above average.

Table 2. List of Aspects and Items in Questionnaire

Aspects	Items	
Intellectual Abilities	1. IQ Score 2. Intelligence 3. Knowledge	4. Academic Grades 5. Language and Writing 6. Mathematics and Sciences
Physical Abilities	1. Physical Activeness 2. Physical Fitness	3. Strength and Coordination 4. Motor Skills
Musical and Artistic Abilities	1. Musical Abilities	2. Artistic Abilities
Emotional/Social Intelligence	1. EQ Score 2. Social Skills 3. Extraversion 4. Likeability 5. Popularity	6. Number of Friends 7. Funniness/Humour 8. Quality of Friendship 9. Sociability
Personal Traits	1. Creativity 2. Confidence 3. Leadership 4. Hardworking 5. Modesty 6. Honesty/Integrity 7. Respectfulness	8. Selflessness 9. Positivity/Optimism 10. Self-awareness 11. Mental Strength 12. Resilience 13. Physical Appearance/Attractiveness
Abstract Aspects	1. Happiness	4. Talent

	2. Religiousness 3. Health	5. Success 6. Overall
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3.2 Pilot Test

Initially, a pilot test was conducted to evaluate the validity of the questionnaire. A total of 6 individuals participated, comprising 4 students and 2 academic supervisors from Kolej PERMATA@Pintar Negara. The participants were provided a link which directed them to the Google Form questionnaire, and were instructed to provide feedback.

During this preliminary testing phase, a number of flaws within our questionnaire were identified, including minor language discrepancies, ambiguously phrased statements and technical issues with the Google Form. As a result, a number of amendments were made before publishing the real questionnaire.

3.3 Participants

A total of 323 individuals of varying backgrounds responded to the questionnaire. The criteria for respondents were Malaysian nationality and an age of 13 or above.

3.4 Procedure & Data Analysis

The questionnaire was distributed via online platforms. Respondents were provided guidelines and reminded to respond as honestly as possible, as there are no definitive “right” or “wrong” answers. Finally, towards the end of the questionnaire, respondents were asked to confirm the validity and accuracy of their provided responses, acknowledging their use for the research purposes. Out of the 323 initial responses received, a total of 318 responses were deemed suitable for inclusion in the final analysis.

The analysis began by converting the Google Form responses into an Excel Spreadsheet so that it could be run in the statistical software using the IBM Statistical Package for Social Sciences (SPSS) Version 29. Descriptive statistics were performed to determine the averages of samples among gifted and non-gifted individuals and between genders. Levene’s Test For Equality of Variances was used to investigate the correlation between self-perception and the previously mentioned factors. General statistical techniques were used to analyse the data with a 95% level of confidence ($\alpha = 0.05$).

4.0 Results

4.1 Demographic Information of Samples

Table 3. Demographic Information of Samples

	Gifted	Non-Gifted	Total
Male	68	52	120

Female	77	121	198
Total	145	173	318

4.2 Overall Mean for Gifted and Non-Gifted

An Independent Sample t-test (Table 4) showed that there is a significant difference in self-perception between gifted and non-gifted individuals. Since the significance level surpasses .05, we reject H_{01} .

Table 4. Overall Mean for Gifted and Non-Gifted

	Giftedness	Mean	Sig.
Overall Mean (across all 40 items)	Gifted	4.44	0.010
	Non-Gifted	4.60	

4.3 Overall Mean for Male and Female

An Independent Sample t-test (Table 5) showed that there is no significant difference in self-perception between males and females. Since the significance level does not surpass .05, we fail to reject H_{02} .

Table 5. Overall Mean for Male and Female

	Gender	Mean	Sig.
Overall Mean (across all 40 items)	Male	4.49	0.843
	Female	4.55	

4.4 Mean and Alpha Values Across All Items

Table 6 compares the means across all items in terms of giftedness and gender, along with its significance level. Significance levels which surpass .05 are in bold.

Table 6. Mean and Alpha Values Across All Items

Aspect	Item	Giftedness	Mean	Sig.	Gender	Mean	Sig.
Intellectual Abilities	IQ Score	Gifted	5.12	0.001	Male	5.02	0.494
		Non-Gifted	4.60		Female	4.73	
	Intelligence	Gifted	4.90	0.007	Male	4.89	0.541
		Non-Gifted	4.72		Female	4.75	
	Knowledge	Gifted	4.86	0.029	Male	4.92	0.315
		Non-Gifted	4.72		Female	4.70	

Aspect	Item	Giftedness	Mean	Sig.	Gender	Mean	Sig.
	Academic Grades	Gifted	4.54	0.074	Male	4.66	0.710
		Non-Gifted	4.74		Female	4.65	
	Language and Writing	Gifted	4.87	0.009	Male	4.69	0.807
		Non-Gifted	4.81		Female	4.92	
	Mathematics	Gifted	4.68	0.071	Male	4.86	0.432
		Non-Gifted	4.62		Female	4.53	
	Sciences	Gifted	4.54	0.028	Male	4.59	0.826
		Non-Gifted	4.58		Female	4.55	
Physical Abilities	Physical Activeness	Gifted	4.10	0.978	Male	4.04	0.020
		Non-Gifted	4.22		Female	4.24	
	Physical Fitness and Strength	Gifted	3.93	0.270	Male	3.94	0.155
		Non-Gifted	4.17		Female	4.13	
	Coordination and Motor Skills	Gifted	4.31	0.096	Male	4.24	0.575
		Non-Gifted	4.33		Female	4.37	
Musical and Artistic Abilities	Musical Abilities	Gifted	3.99	0.041	Male	3.39	0.006
		Non-Gifted	3.36		Female	3.81	
	Artistic Abilities	Gifted	4.08	0.042	Male	3.30	0.069
		Non-Gifted	3.58		Female	4.11	
Emotional/Social Intelligence	EQ Score	Gifted	4.52	0.942	Male	4.34	0.473
		Non-Gifted	4.50		Female	4.61	
	Social Skills	Gifted	4.06	0.250	Male	4.27	0.556
		Non-Gifted	4.55		Female	4.36	
	Extraversion	Gifted	3.89	0.643	Male	4.00	0.385
		Non-Gifted	4.20		Female	4.09	
	Likeability	Gifted	4.45	0.653	Male	4.58	0.920
		Non-Gifted	4.75		Female	4.63	
	Popularity	Gifted	3.94	0.806	Male	4.25	0.943
		Non-Gifted	4.37		Female	4.13	
	Number of Friends	Gifted	4.21	0.677	Male	4.28	0.496
		Non-Gifted	4.47		Female	4.40	
	Funniness/Humour	Gifted	4.61	0.123	Male	4.40	0.389
		Non-Gifted	4.73		Female	4.84	
	Quality of Friendship	Gifted	4.89	0.082	Male	4.99	0.082
		Non-Gifted	5.14		Female	5.05	
	Sociability	Gifted	4.17	0.107	Male	4.43	0.733
		Non-Gifted	4.67		Female	4.45	
Personal Traits	Creativity	Gifted	4.70	0.867	Male	4.56	0.542
		Non-Gifted	4.62		Female	4.71	
	Confidence	Gifted	4.10	0.361	Male	4.44	0.566
		Non-Gifted	4.68		Female	4.40	
	Leadership	Gifted	4.32	0.330	Male	4.53	0.685
		Non-Gifted	4.70		Female	4.53	
	Hardworking	Gifted	4.31	0.250	Male	4.45	0.937
		Non-Gifted	4.82		Female	4.67	
	Modesty	Gifted	4.57	0.195	Male	4.64	0.357
		Non-Gifted	4.71		Female	4.65	

Aspect	Item	Giftedness	Mean	Sig.	Gender	Mean	Sig.
	Honesty/ Integrity	Gifted	4.64	0.646	Male	4.98	0.079
		Non-Gifted	5.21		Female	4.93	
	Respectfulness	Gifted	5.10	0.055	Male	5.29	0.562
		Non-Gifted	5.30		Female	5.16	
	Selflessness	Gifted	4.59	0.199	Male	4.68	0.281
		Non-Gifted	4.88		Female	4.79	
	Positivity/ Optimism	Gifted	4.46	0.762	Male	4.65	0.361
		Non-Gifted	4.79		Female	4.64	
	Self-awareness	Gifted	4.99	0.085	Male	4.94	0.159
		Non-Gifted	5.02		Female	5.05	
	Mental Strength	Gifted	4.52	0.436	Male	4.65	0.837
		Non-Gifted	4.76		Female	4.65	
	Resilience	Gifted	4.68	0.564	Male	4.88	0.422
		Non-Gifted	4.89		Female	4.74	
	Physical Appearance/ Attractiveness	Gifted	3.84	0.747	Male	3.92	0.486
		Non-Gifted	4.34		Female	4.23	
Abstract Aspects	Happiness	Gifted	4.43	0.004	Male	4.68	0.240
		Non-Gifted	4.91		Female	4.70	
	Religiousness	Gifted	4.08	0.769	Male	4.20	0.749
		Non-Gifted	4.42		Female	4.30	
	Health	Gifted	4.39	0.620	Male	4.65	0.255
		Non-Gifted	4.71		Female	4.51	
	Talent	Gifted	4.48	0.650	Male	4.68	0.463
		Non-Gifted	4.59		Female	4.46	
	Success	Gifted	4.46	0.840	Male	4.52	0.720
		Non-Gifted	4.62		Female	4.57	
	Overall	Gifted	4.72	0.144	Male	4.86	0.324
		Non-Gifted	4.94		Female	4.83	

4.5 Distribution of Rankings for Overall Self-Assessment

Figure 1 shows the frequency of responses for each ranking option when participants were asked to rate themselves in an overall sense. The bar charts represent the responses among all participants, females, males, gifted individuals and non-gifted individuals respectively.

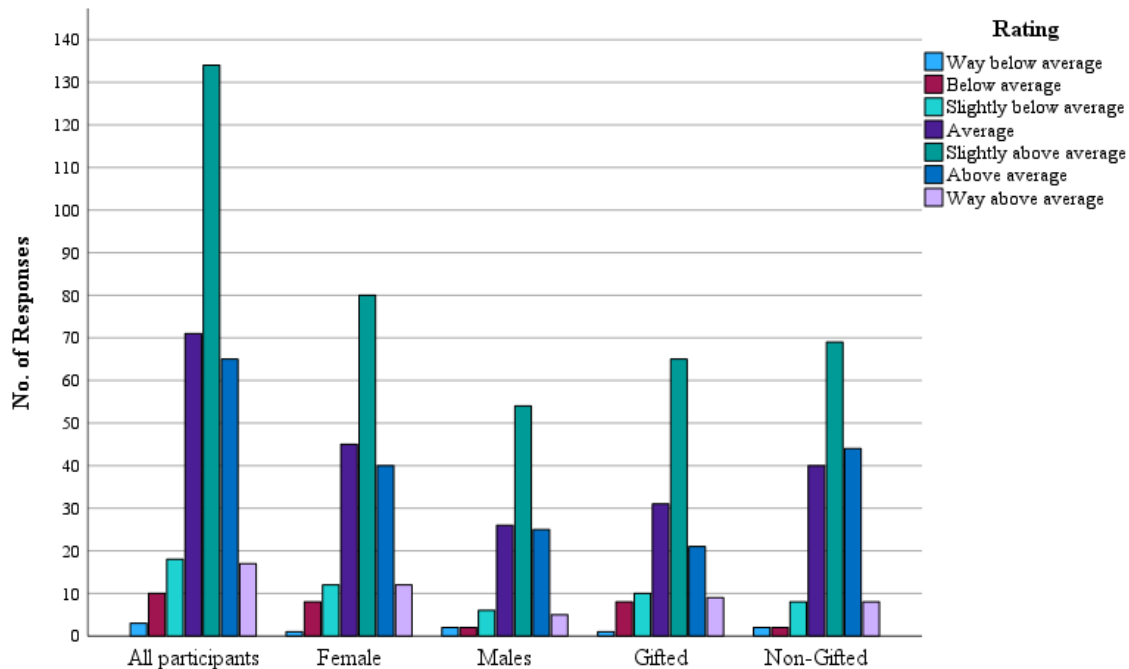


Figure 1. Distribution of rankings for overall self-assessment

5.0 Data Analysis and Discussion

As outlined in Table 4, the results of this study provide conclusive evidence that non-gifted individuals (mean of 4.60) generally have higher levels of self-perception than gifted students (mean of 4.44), with a significance level of 0.01. On the other hand, the same cannot be concluded for self-perception between males and females, as shown in Table 5.

Additionally, the analysis of the results in Table 6 yielded multiple statistically significant findings with a significance level below 0.05. Among them, it is evident that gifted individuals exhibited higher perceptions of various intellectual abilities compared to their non-gifted counterparts, in line with previous research (Albright, N.M., 1965; Hoge, R.D. & Renzulli, J.S., 1993). Specifically, gifted individuals reported a higher perception of their IQ (mean of 5.12) in comparison to non-gifted individuals (mean of 4.6). Similarly, in terms of perceived intelligence, gifted individuals (mean of 4.9) surpassed non-gifted individuals (mean of 4.72) and in the domains of language and writing skills, gifted individuals (mean of 4.87) also displayed a higher perception compared to non-gifted individuals (mean of 4.81). Interestingly, however, non-gifted individuals (mean of 4.58) had a higher perception of their science proficiency than gifted individuals (mean of 4.54), deviating from the general trend.

Gifted individuals also demonstrated higher perceptions in their musical (mean of 3.99) and artistic abilities (mean of 4.08) compared to non-gifted individuals (mean of 3.36 and mean of 3.58, respectively). Notably, another highly significant finding is that non-gifted individuals

perceive themselves as happier (mean of 4.91) than gifted individuals (mean of 4.43), with a significance level of 0.004.

When examining self-perception across gender, females displayed a higher perception of their physical activeness (mean of 4.24) in contrast to males (mean of 4.04). Also, females exhibited a higher perception of their musical abilities (mean of 3.81) compared to males (mean of 3.39). Nonetheless, as with previous studies (Hoge, R.D. & Renzulli, J.S., 1993), gender failed to provide significant results in most areas.

Furthermore, Figure 1 illustrates normal distributions with a rightward skew. Similar patterns were consistently observed with many other aspects too, including happiness, health, talent, success and many more. This common trend could be an indication of a tendency for individuals to perceive themselves more positively than objectively warranted (Robins & John, 1994).

The Cronbach's alpha coefficient for the questionnaire yielded a strong internal consistency of 0.953, indicating that the questionnaire is likely to produce consistent and dependable results.

6.0 Limitations

The uneven distribution of sample sizes across the various groups presents a challenge in drawing direct comparisons and conclusions. This imbalance might affect the statistical power and robustness of the analyses, potentially skewing the representation of self-perception patterns within specific subgroups. In light of this, future research endeavours in this domain should consider balancing sample sizes across genders and groups to ensure a more equitable representation of perspectives.

7.0 Conclusion

In conclusion, this research primarily aimed to study if giftedness affects self-perception across various aspects, while also gaining insights into the impact of gender. Through a comprehensive analysis of the collected data, several noteworthy findings emerged. Notably, the findings indicated that non-gifted individuals generally exhibit higher levels of self-perception than their gifted counterparts. This intriguing revelation prompts further exploration into the underlying factors that contribute to this phenomenon. The identified findings also extend to specific domains such as intellectual abilities, physical abilities and emotional/social intelligence, providing a nuanced understanding of self-perception dynamics. Moreover, several observations were made regarding gender-related differences in self-perception. Ultimately, this study contributes to the broader discourse on self-perception, shedding light on its relationship with giftedness and gender. As society continues to evolve, an in-depth comprehension of

self-perception is invaluable for fostering healthy development, guiding educational strategies, and supporting individuals in reaching their full potential.

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